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Breaking Barriers: Unlocking the Potential of the Offshore Victoria HPHT Tight Gas Discovery

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Abstract

Hydraulic fracturing campaigns on the Norwegian Continental Shelf (NCS) are most commonly executed in chalk formations, while numerous tight sandstone gas opportunities remain unexploited. Historically, technical and operational challenges have included achieving an environmentally compliant fluid for HPHT conditions, the relatively low efficiency of offshore hydraulic fracturing operations, and reluctance to rig up complex coiled tubing spreads on expensive semi-submersible rigs.

Victoria is one of these discoveries, with pore pressures exceeding 800 bar and temperatures reaching 200°C, containing an estimated gas initially in place (GIIP) of 140 BCM. Discovered in 2000, but difficulty in identifying a commercially viable development solution resulted in the discovery being relinquished in 2018. However, advancements in oilfield technology and completion techniques now offer solutions to the challenges previously faced.

Detailed geomechanics analyses of offset wells and fine-grid 3D fracture simulations were conducted to assess hydraulic fracture feasibility and potential fracture propagation in tight gas units. Results suggest that fractures could be successfully placed, covering up to 70% of the main target unit. Modern multi-stage completion technologies were modelled to confirm that hydraulic fracturing operations could be executed within typical maximum working pressures of fracturing and completion equipment. Efficiency improvements for a multi-stage campaign were considered, particularly the use of drill pipe manipulated sleeves for rapid screen-out recovery, eliminating the need for complex coiled tubing rig-ups. A qualitative assessment of fracturing fluid options was performed. Finally, reservoir simulation incorporating the resulting fractures was carried out to evaluate production potential for multi-well field development.

This paper details the integration of the latest stimulation technologies, transforming these tight gas discoveries from complex and uneconomic to commercially viable and highly workable opportunities. In Victoria, reservoir simulation results predicted a recovery factor of 29%, or about 40 BCM, based on a six-well development scenario. The feasibility of multi-stage stimulation using drill pipe manipulated sleeves removes the necessity for complex coiled tubing rig-ups, significantly reducing campaign costs and operational complexity. These findings demonstrate that an economic tight gas development of the Victoria discovery is now viable with modern stimulation and completion technologies.

On the NCS there are a further 7 major unlicensed tight gas discoveries. The methodologies and technological applications presented here provide a blueprint for unlocking these and other tight gas opportunities on the NCS.

Background

The Victoria discovery is located in the Norwegian Sea in block 6506, discovered in 2000 by ExxonMobil E&P Norway. It is a high pressure high-temperature (HPHT) discovery with crestal structure, pore pressures of around 800 bar, and reservoir temperatures reaching 200°C. Water depth is 420 m. The gas bearing sands are Jurassic, consisting of the Garn, Ile and Tilje sands, which are fairly low permeability sandstones requiring hydraulic fracturing to be economically viable. The crestal depth is at 4820 m TVDSS with gross reservoir thickness of about 260 m and a gas column of about 450 m. Victoria also has CO₂ content up to 10%. Figure 1 shows the geographic location of Victoria.

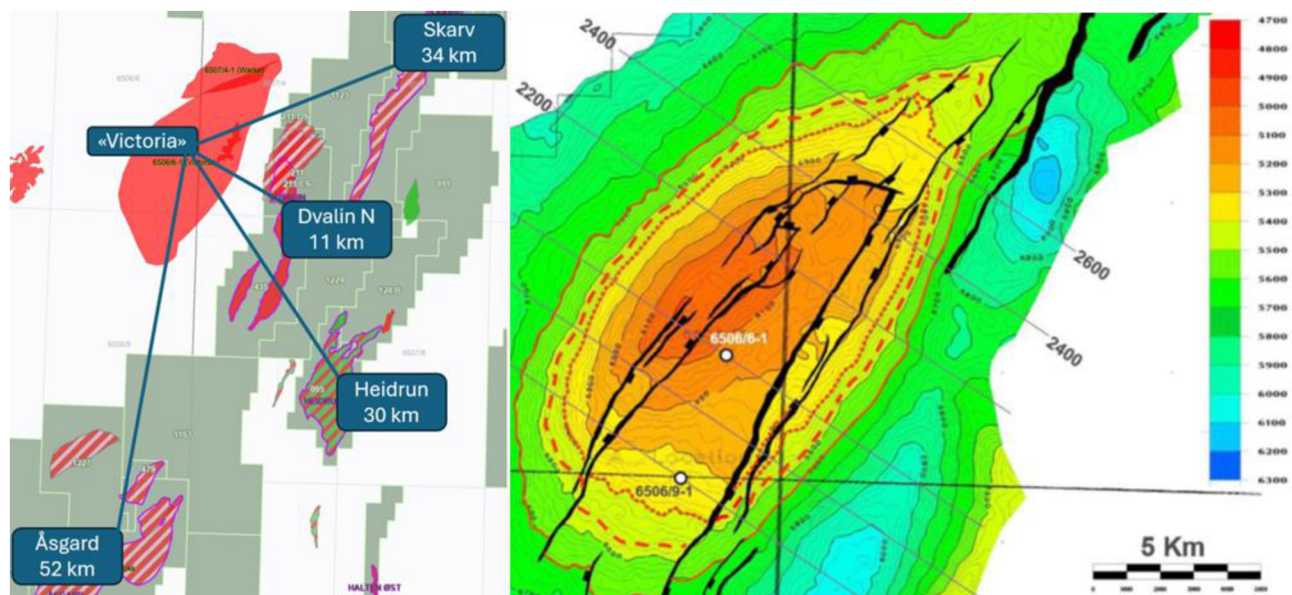


Figure 1—Victoria location on the NCS (left), and the Victoria structure with the discovery and appraisal wells.

The discovery well, 6506/6-1, was drilled at the crest of the structure, also shown in Figure 1. During the drilling, testing and logging operations, there were some issues encountered – the Ile and Tilje sands were found to be extremely hard, a leak in the liner resulted in the drill stem test (DST) to be abandoned, and the logging tools failed due to the high temperature. Years later, in 2009, a new operator TotalEnergies EP Norge, drilled an appraisal well, 6506/9-1, towards the edge of the structure, south of the discovery well. Technology advancements since the drilling of the first well meant that a similar drilling, testing and logging programme could be carried out with minimal issues, resulting in a set of good quality data, and an additional mini-frac injection test.

While hydraulic fracturing would have been identified as a potential development solution for Victoria at the time, multiple challenges would have to be overcome. Firstly, the high temperatures of Victoria would, without question, require a high performance fracturing fluid, and given Norway's stringent environmental regulations, would be dependent on extensive testing of all proposed chemicals or possibly even development of new chemicals with no guarantee that they would be compliant. Besides potential issues with being able to mature a solution for a suitable fracturing fluid system there would have been several other concerns at the time. The availability of stimulation vessels would have been a concern, with the market being very competitive in the late 2000's and early 2010's, with exposure to having an expensive

semi-submersible waiting on a vessel. Further apprehension existed in the industry around the rigging up of a Coiled Tubing (CT) spread on a semi-submersible in case of a screen-out, from both well control and rig safety perspectives. This was regarded as a high-risk operation at the time, having been done a limited number of times in the North Sea region. Additionally, the possibility of a screen-out and an extensive clean-up operation would have posed a further economical challenge. Finally, given the available technology at the time, only short fractures of around 10-20 m in fracture half-length would be achievable, where in fact half-lengths of closer to 100 m are required for optimum productivity. While other design and operations solutions could be met with time, such as appropriately pressured-rated completions, the aforementioned difficulties would have appeared more like barriers. No further wells were drilled in Victoria and the license was eventually relinquished in 2018.

Almost two decades later, the technological developments and improvements in both technical and operational domains suggest that the effective development of Victoria in an operationally efficient manner is not just possible, but realistically achievable. In addition to Victoria there are numerous other gas discoveries on the NCS that have been relinquished over the years. Currently there are 8 major unlicensed discoveries, shown in Figure 2, each with GIIP greater than 15 BCM, containing an estimated total of 650 BCM of gas. These face a variety of challenges to development, but a common feature is that they are tight sandstones where hydraulic fracturing can be a viable technology.

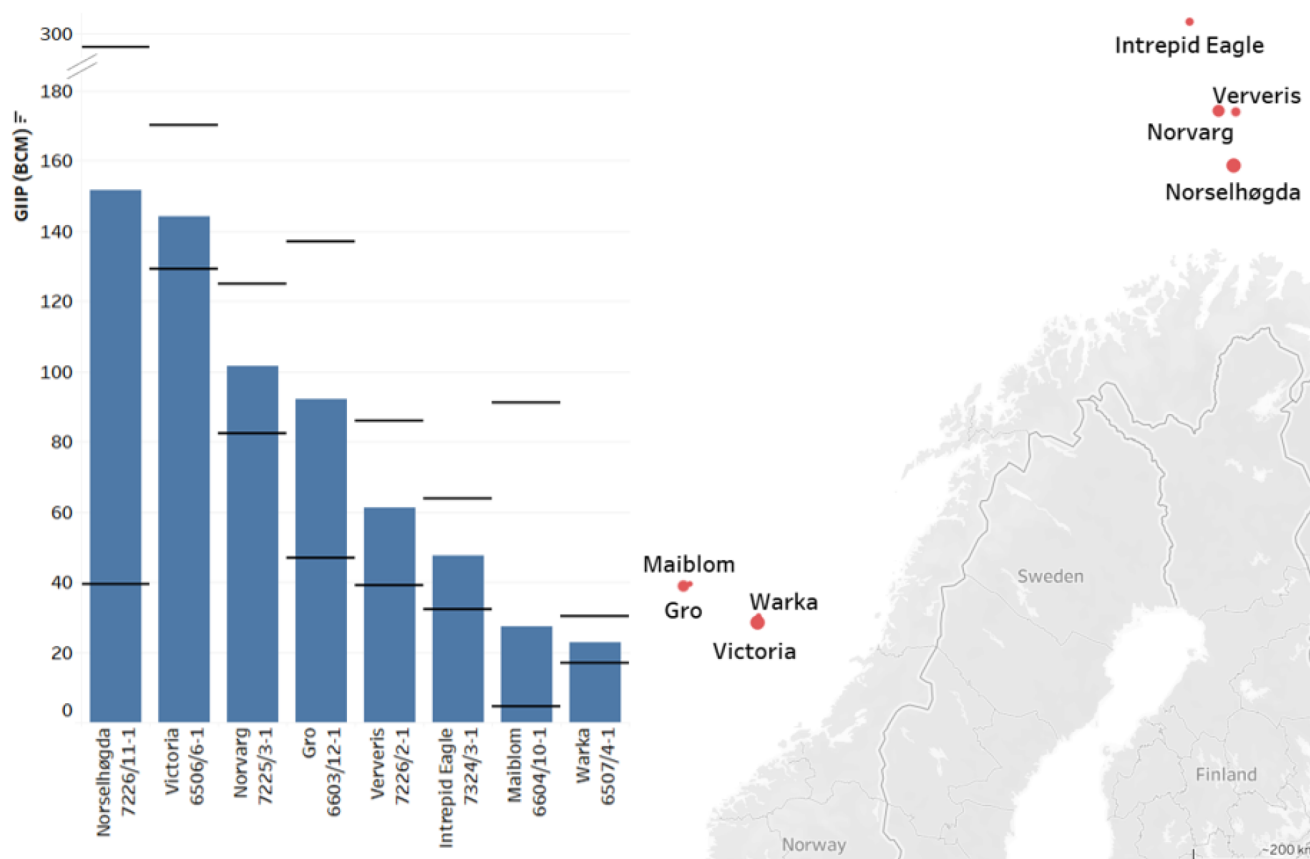


Figure 2—GIIP estimates (lines indicate low and high values) for major unlicensed gas discoveries (name and discovery well) on the NCS.

Fracture Geomechanics and Simulations

A review of the fracturing geomechanics was essential to carry out, to assess whether a hydraulic fracture could be placed with resulting properties that would indeed translate Victoria into a commercially workable opportunity. A full review of the available data from the two wells (6506/6-1 and 6506/9-1) revealed that

a full set of data was available for well 6506/9-1, including sonic logs that are critical for calculating rock mechanics profiles. Furthermore, the mini-frac data for the same well would provide a stress calibration point with which to align the minimum horizontal stress profile. Ultimately, a detailed geomechanics model with fine vertical layering was developed, though due to the lack of core rock mechanics data, the static Young's Modulus and Poisson's Ratio were estimated from regional correlations. It should be noted, though, that core for well 6506/9-1 was taken, and so future development should consider completion rock mechanics testing on this core. A cross-check of Gamma Ray logs for both 6506/9-1 and 6506/6-1 showed that similar features and reservoir quality could be correlated between the wells, and therefore the use of 6506/9-1, located away from the crest of the structure, was suitable to use as a representative offset well.

The local fracture gradient provided from the small mini-frac injection was equivalent to 0.183 bar/m (985 bar at 5,387 m TVDSS), which would likely result in very high surface treating pressures. Aligning the stress profile to this fracture gradient at this depth, resulted in a stress profile with relatively high stress contrast between adjacent layers, suggesting that some layers may act as barriers to fracture height growth, allowing hydraulic fractures to develop longer, and more contained fractures. The fluid leak-off coefficient profile was inferred from the permeability profile, with the leak-off coefficient averaging between 0.002 – 0.003 ft/sqrt.min, a fairly typical leak-off coefficient observed in hydraulically fractured wells in this region.

The Ile and Tilje layers were identified as the main targets in Victoria. Fracture simulations for a horizontal well in both units were carried out, testing fracture initiation from the lowest stressed layers. A conservative pump schedule with around 50% pad volume was simulated to ensure the net pressures did not exceed around 100 bar, using fluid rheology properties of a representative crosslinked synthetic polymer gel, and 16/30 resin coated, high strength proppant (HSP), suitable for the high stress environment. A gentle proppant schedule was applied, simulating placement of a range of proppant volumes between 125 – 225 metric tons (MT) of proppant.

The Ile formation shows a region of tight sands (therefore lower leak-off coefficient) with increased stress in the centre of the unit, which prevents the fracture height from covering the full unit. Initiation in the Upper Ile showed excessive height growth into the overlying non-net Not, and Garn formations, with most of the conductivity being placed above the wellbore. When the fracture is initiated in a low stressed layer in the Lower Ile, the simulation showed that the hydraulic fracture did cover approximately 60% of the interval, achieving around 120 m propped fracture half-length, and an average conductivity of 90 mD.m, with high conductivity at the wellbore. However, there was some downward fracture growth observed into the non-gas bearing Ror formation. Fractures profiles for initiation in the Ile unit are displayed in Figure 3.

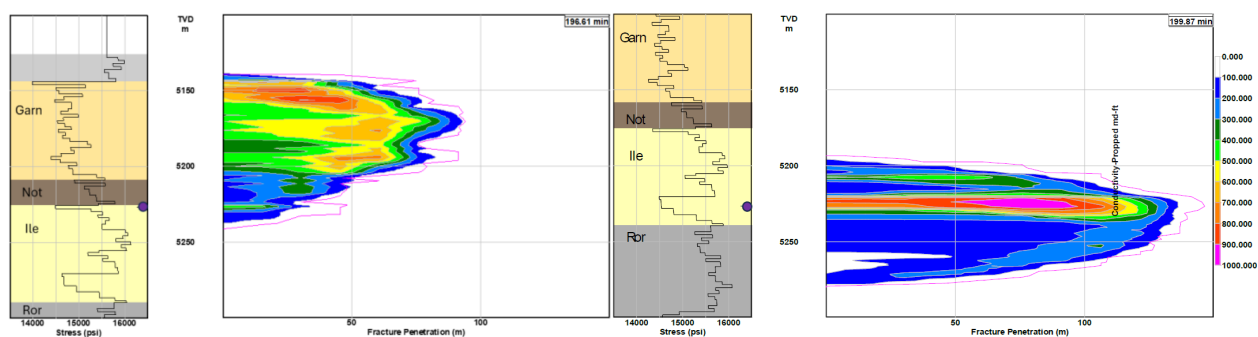


Figure 3—Fracture simulations in Ile formation, initiated in the Upper Ile (left), and the Lower Ile (right).

The minimum horizontal stress profile for the Tilje formation also showed some character and reasonable stress contrast. Once more, two initiation depths were tested. Fractures initiated in the Tilje, shown in Figure 4, are confined, covering up to 70% of the target, and no growth into the units above or below the target was observed. Propped fracture half-lengths around 120 m were achieved, with average conductivities of

130 mD.m and high conductivity in the near wellbore region. A combination of fractures initiated in low stressed layers in both Upper and Lower Tilje would be optimal for maximising reservoir contact.

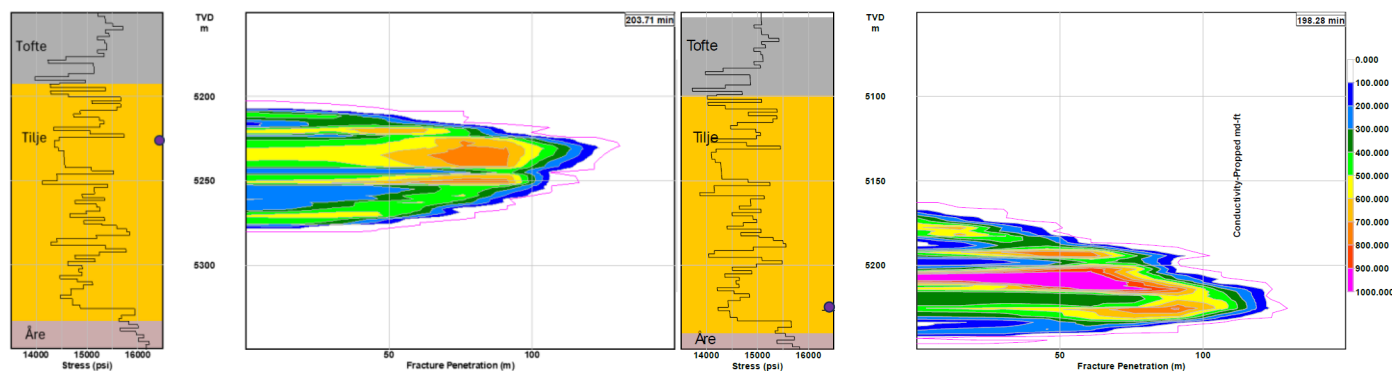


Figure 4—Fracture simulations in the Tilje formation, initiated in the Upper Tilje (left), and Lower Tilje (right).

Fracturing Fluids

Fracturing fluids are arguably the most important part of the fracturing design. Sufficient viscosity is required to transport and suspend the proppant, while the crosslinking mechanism must be shear recoverable, after exposure to high shear within the tubulars, or else start to crosslink once within the fracture. Stability times are important so the fluid does not break prior to the treatment ending, as is the breaking mechanism itself for enhanced fluid flowback. Typical fracturing fluid systems used for fracturing sandstones in the North Sea, for a long time, have been guar or derivatized guar based fluids, crosslinked with borate and/or zirconate crosslinkers. Stable fluids can generally be developed for temperatures up to 150°C, however the performance drastically diminishes at higher temperatures than this, with very short stability times. Considerable pre-job cooldown injections would be required prior to introducing proppant to the formation, which becomes time consuming, expensive, and can also be logistically complex.

Alternative fracturing fluids systems that can perform in higher temperatures around 200°C have been used in the industry and documented from around the early 2010's, which are synthetic polymer based, and crosslinked with organometallic crosslinkers. These fluid systems have enhanced stability at ultrahigh temperatures, as well as a series of additional benefits, outlined by [Christiawan et al \(2016\)](#), including the much lower fluid friction pressures, control over the breaking mechanism, fast hydration times and the fact that the polymer is not degraded by the presence of bacteria, removing the need for biocide, which is usually a restricted chemical in Norwegian waters. Such fluid systems were not in use during the original feasibility reviews, but now provide a solution for Victoria, and indeed other HPHT discoveries in the region.

Completion Options

At the time of the original development screening, the most common method of multi-stage fracturing completion was the 'plug and perf' method, that requires at least a single CT intervention between each fracture treatment to isolate the stages. Furthermore, a CT run to TD is required thereafter, to remove the selected plug, whether it be a sand or a mechanical plug, and clean the well. Economically, when hydraulically fracturing from a semi-submersible rig on location, this completion method was unattractive.

In the early 2010's, a transition was made from 'plug and perf' to ball actuated sleeve completions, which removed the requirement for CT intervention between stages. [Langford et al \(2013\)](#) demonstrates that if the weather offshore is favourable, and no screen-out events occur, hydraulic fracture treatments can be executed and at a rate of one hydraulic fracture per day, including injection test schedules, provided there is no screen-out event. [Channa et al \(2023\)](#) further details ball dropping procedures adapted to conventional

reservoirs, to reduce the probability of requiring CT intervention between stages. However, the risk of screen-out itself adds a significant number of days, and therefore large costs, to a campaign.

The most recent and successfully implemented multi-stage completion method in the North Sea region is a drill-pipe manipulated sleeve method. The lower completion consists of numerous sleeves, shifted open using a specific tool on drill string prior to fracturing. A packer is set below the sleeve, to ensure isolation from the previous stage. The hydraulic fracture treatment is then pumped via the drill string through the open sleeve. After fracturing, the sleeve is shifted closed, and the contents of the drill pipe (usually a small volume of proppant laden fluid) is reverse circulated out, before tripping pipe to the next sleeve. All sleeves can be opened using a tractor after the fracturing operations. The benefits of this system are numerous:

- No Coiled Tubing is required.
- All fractures can be executed in a single trip, and fractures don't necessarily need to be executed sequentially, for example, from toe to heel.
- The pressure via the annulus provides live downhole pressure responses, the availability of which will contribute to better decision making during the treatment, without having friction masking the fracturing pressure trends.
- Perhaps most critical of all, recovery from a screen-out event is almost operationally identical to normal treatment execution (no screen-out), aside from directing proppant laden fluid reverse circulated from within the drill string into dedicated tanks or skips. This is a substantial saving on days and therefore cost.

Jøranson et al (2024) discuss the achievement of executing multiple fractures per day in offshore Norway with this multi-stage hydraulic fracturing technique, a game changer for HPHT wells in the deeper waters on the NCS.

Single Trip Multi Frac Method Verification

The high minimum horizontal stress magnitude results in high treating pressures. Single-trip multi-frac type completions such as drill pipe manipulated sleeves typically use drill pipe sizes 2-7/8" to 3-1/2" for the lower completion, which causes some very high fluid friction, resulting in even higher surface treating pressures. Tentative calculations were made, using friction factors obtained from a series of recent campaigns executed with the same completion technique, to verify that a drill pipe system would be suitable for Victoria. Surface injection pressures could indeed be below the standard 1000 bar limit for hydraulic fracturing equipment, if a slightly weighted fluid was used, and wellhead equipment (seeing the drill pipe-tubing annulus pressures) was appropriately pressure-rated. Use of a synthetic polymer based fluid discussed previously, would reduce surface injection pressures even further, to be well within typical limits of the service equipment.

Reservoir Modelling

The newest completion methods and development in fracturing fluids systems suggest that hydraulic fracturing in Victoria is operationally feasible. Reservoir modelling was carried out to determine whether the achievable hydraulic fracturing parameters would provide sufficient uplift in production and therefore be a development opportunity. The model of Victoria has a GIIP of 137 BCM, with 30 BCM in the Ile formation and the rest in Tilje. The various identified facies have been lumped into good and tight sands, the areal distribution of which are shown in Figure 5, with the GIIP distribution by permeability classes for both. The good sands make up 20% of the rock volume, 50% of the pore volume and almost 70% of the GIIP. Vertical connectivity is assumed very poor, with vertical to horizontal permeability ratio (K_v/K_h) of 0.1 in good sands and 0 in tight sands. With the good sands typically being quite thin (less than 5 m), hydraulic fractures are important for vertically connecting them across the poor tight sands.

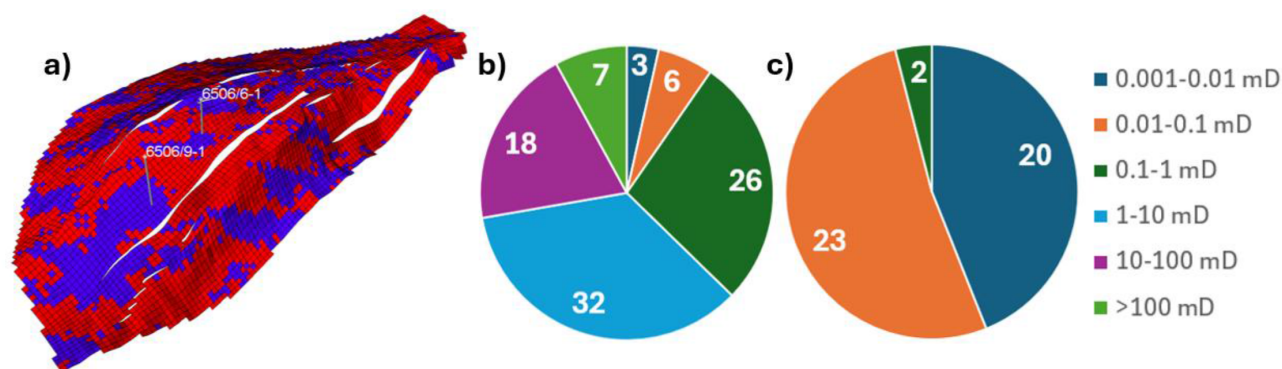


Figure 5—Areal distribution of good (red) sands in Upper Ile (a) along with GIIP distribution (BCM) by permeability classes for good sands (b) and tight sands (c) across the entire model.

It was assumed that highly deviated wells with 2000 m reservoir sections, targeting all formations are feasible. Wells were modelled in the direction of the minimum horizontal stress (SW-NE), in line with local stress data, so that fractures would be transverse and thus increase reservoir contact. Each well was modelled with 9 transverse fractures, 1 targeting initiation in Lower Ile, 4 in Upper Tilje and 4 in Lower Tilje. The good sands in Upper Ile have thickness of 2-4 m and sufficient permeability that regular perforations should suffice. The hydraulic fractures were explicitly modelled in the simulator with local grid refinements (LGR). The properties from fracture simulations were input into the model, using around 75% of the propped half-length as the effective half-length, that is, the fraction of the half-length contributing to production, and the average conductivities as simulated, which already have a high fluid damage factor applied. This corresponds to half lengths of about 100 m and 130 mD.m conductivity. To prevent excessive numerical convergence problems the fractures were modelled as 0.3 m wide with 430 mD permeability. As comparison, wells without hydraulic fractures were also modelled. These had similar length reservoir sections and followed the same SW-NE direction. Vertically, however, they were somewhat optimized to follow layers with a higher proportion of good sands but still penetrating the complete thickness.

Production schedules were run for 30 years, with a 10 million Sm³/d plateau rate and a tubing head pressure limit of 90 bar. Figure 6 shows the development scenarios where up to 7 wells were tested with a 6 well scenario chosen as the base case since the 7th well provided comparatively little additional production. Individual well recoveries varied between 5 and 9 BCM. The no-fracturing case with 6 wells resulted in 31 BCM cumulative production, highlighting the substantial benefit that hydraulic fracturing provides over regular open hole completions. The additional production in the fracturing case is mainly achieved by extending the plateau period, 5 versus 3 years, making it comparatively more economically attractive. The fracturing case is also more robust with respect to vertical well placement, with the no-fracturing case reliant on the wellbores intercepting a number of the good sands. Using the same vertical well path as in the fracturing case, would reduce recovery to 18 BCM.

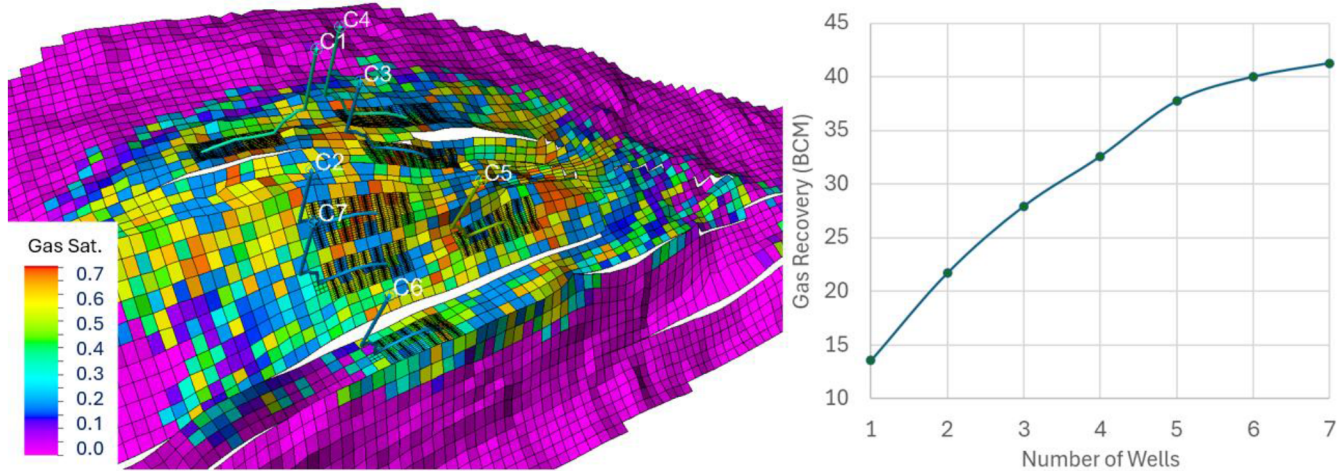


Figure 6—Well placement along with cumulative gas production for development scenarios up to 7 wells with hydraulic fracturing.

Recovery from Ile in the 6 well scenario, about 7 BCM, representing 22% recovery factor from Ile, was comparatively similar with or without fractures, suggesting the good sands towards the top are dominating recovery. The Tilje formation is dominated by thinner sands, distributed throughout the section. Connecting these with the hydraulic fractures, improved recovery substantially over the non-fractured case, resulting in a 31 versus 23% recovery factor from Tilje, corresponding to 33 and 24 BCM respectively, shown in Figure 7. In terms of recovery from good versus tight sands, 32 BCM of the recovery in the fractured case came from the good sands, contributing 80% of the overall recovered gas. However, this is still only a 34% recovery factor from these sands. The overall field recovery factor was 29 and 23% for the fractured and non-fractured cases respectively.

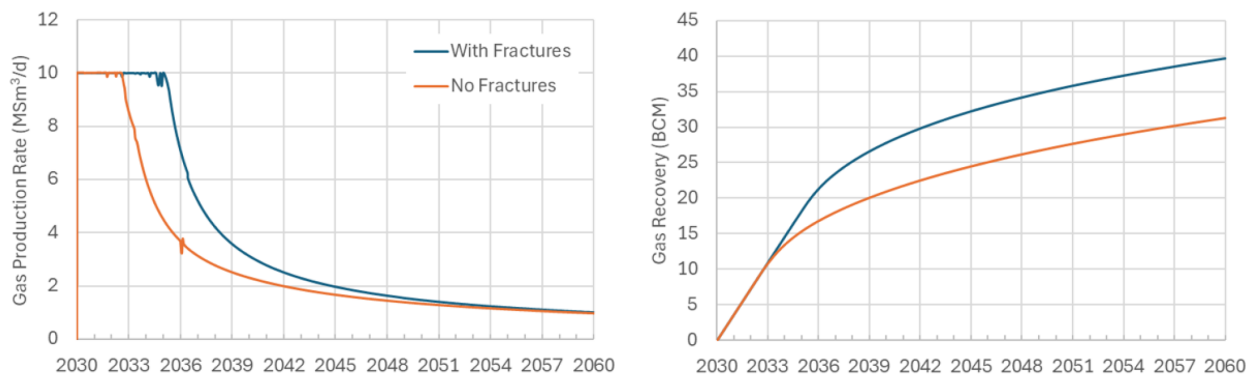


Figure 7—Field gas production rate and cumulative recovery for a 6 well development scenario, comparing scenarios with and without hydraulic fractures.

Some sensitivities were done on the 6 well fractured case to assess the robustness, shown in Figure 8:

- Permeability multipliers of 0.5 and 1.5 were applied to all cells. This was found to have quite significant impact, with recovery varying between 32 and 44 BCM as a result.
- The base well configurations have about 2000 m reservoir sections, completed in both Ile and Tilje, with 9 transverse fractures. As a more conservative case the wells were shortened to 1500 m with 7 fractures, completed in Tilje only. This reduced recovery to 32 BCM, mainly due to recovery from Ile being almost 0. Another possibility could be to combine 5 fractured wells in Tilje with a single non-fractured well in Ile. This would increase recovery to 34 BCM.

- Vertical connectivity is already assumed to be quite poor, with 0 K_v/K_h outside the good sands. Reducing K_v/K_h in good sands to 0.01 (from 0.1) only reduces recovery by 1 BCM. There is, however, an upside potential if vertical connectivity is better than assumed. A K_v/K_h of 0.01 for both good and tight sands increase recovery to 47 BCM.
- In a 6 well scenario, wells are placed in all major fault blocks and fault transmissibility does not have a major impact. Reducing transmissibility factor from a base case of 0.5 to 0.01 reduces recovery to 36 BCM, whereas open faults only gain less than 1 BCM.
- The main advantage of the fractures is vertically connecting the good sands that are typically quite thin. Reducing fracture half length from 100 m to about 65 m only have a moderate impact, reducing recovery to 38 BCM.

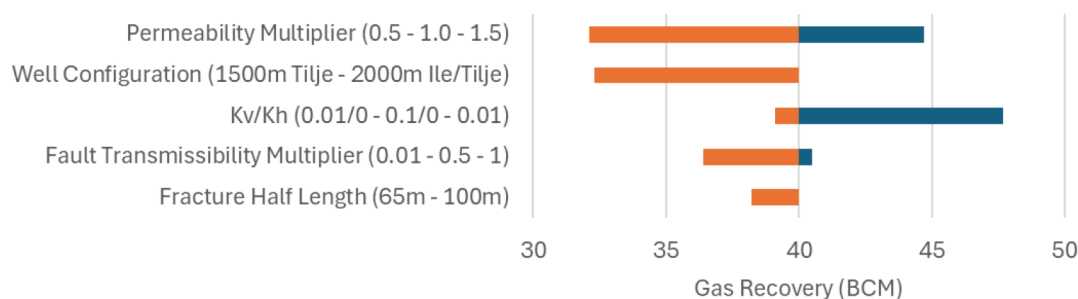


Figure 8—Investigated sensitivities, along with values for low-base-high, for a 6 well scenario with hydraulic fractures.

Conclusion

Detailed geomechanics modelling and hydraulic fracture simulation reveals that fracture properties required for productivity can be achieved. At the same time, it is the most recent developments in multi-stage hydraulic fracturing completions and fracturing fluid systems have helped development opportunities for Victoria to become commercially attractive. Fracturing fluid systems for ultra high temperature reservoirs now exist and have been successfully used in other markets. However, they will still need to be approved for use in Norwegian waters. Perhaps the most important technology to allow efficient hydraulic fracturing operations in HPHT fields such as Victoria, is the single trip multi frac type completion using drill pipe to manipulate sleeves, that allow for screen-out recovery to be fast, and completely remove the requirement for coiled tubing on a semi-submersible. These two critical advancements form the blueprint for unlocking the potential at Victoria, and the several other tight gas development opportunities on the NCS holding sizeable gas in place. While there still exists challenges for successful development of the Victoria discovery, 40 BCM gas recovery based on a 6 well scenario should warrant operators to revisit this discovery.

Acknowledgements

The Norwegian Offshore Directorate would like to thank TotalEnergies EP Norge for sharing the dynamic model used during the original assessment of the Victoria discovery.

Abbreviations

BCM	Billion Cubic Metres
CT	Coiled Tubing
DST	Drill Stem Test
GIIP	Gas Initially in Place
HPHT	High Pressure High Temperature
K_v/K_h	Vertical to Horizontal Permeability Ratio
LGR	Local Grid Refinement

NCS Norwegian Continental Shelf
TD Total Depth
TVDSS True Vertical Depth Subsea

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